DC Scattering time in graphene from mobility measurements

We use the mobility measurements in Electric Field Effect in Atomically Thin Carbon Films (https://www.science.org/doi/full/10.1126/science.1102896?casa_token=UF3_Ns1F8TYAAAAA%3ADXkGRq7yLL5srhyEJP2sg8aRSca13Gw0pRpcA0be1aeJdm3ui6_be3NuLJQqV-3OE72ICOZGvbVWmTQ4) in order to find the DC scattering time in graphene. From the paper cited above, we know that the mobility, μ and the DC conductivity, σ are related as follows:

$$\sigma = \lim_{\omega \to 0} \frac{ie^2 E_F}{\pi \hbar^2 (\omega + i/\tau)} = \frac{e^2 E_F \tau}{\pi \hbar^2} = ne\mu$$

Furthermore, we note that we have $n = g_s g_v \pi k_F^2/(4\pi^2)$, where $g_s = g_v = 2$ are the spin and valley degeneracies, respectively. Using the linear dispersion of graphene, i.e. $E_F = \hbar v_F k_F$, we may write the electronic density in terms of the Fermi energy as follows:

$$n = E_F^2/(\hbar^2 v_F^2 \pi) \leftrightarrow E_F = \hbar v_F \sqrt{\pi n}$$

Therefore, writing the decay time in terms of the electronic density gives us:

$$\tau = \pi \hbar^2 n\mu/eE_F = \hbar\sqrt{\pi n}\mu/ev_F,$$

which is the result cited in Plasmonics in graphene at infrared frequencies (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.80.245435?casa_token=QoZK8jI_kgIAAAAA%3A41U2yqebBFQC2mDR4Vxzy3WVE8xxu8OiAQx3W_DQPLobNfexjIxrSqtc81qqHzFORM3JjIZmrSHre8St). We now use the values $n = 10^{13}/\text{cm}^2$ and $\mu = 10,000 \text{ cm}^2/(\text{Volt} \times \text{s})$, $v_F = 10^8 \text{cm/s}$, we obtain:

$$\begin{aligned}\tau = 1.05 \times 10^{-34} \text{Joules} \times \text{s}\sqrt{\pi 10^{13}} \times 10000/(1.6 \times 10^{-19} \text{Joules} \times 10^8) = \\ 1.05 \times 10^{-13} \text{Joules} \times \text{s}\sqrt{\pi 10}/(1.6 \times \text{Joules}) = 3.7 \times 10^{-13}\text{s}\end{aligned}$$

This is not quite $6.4 \times 10^{-13}\text{seconds}$ as cited in the Soljacic paper. However, in that case, they took $n = 3 \times 10^{13}/\text{cm}^2$, so we need to multiply the answer above by $\sqrt{3}$, which give $6.37 \times 10^{-13}\text{seconds}$, as desired.